VERSION
14.0

AMS GRA ARCHITECTURE VOLUMES

Processor Chassis and Backplane (PCaB) MPU

AFLCMC/WA

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1.0 Overview

A Processor Chassis and Backplane (PCaB) is a Level 3 class Minimum Procurable Unit (MPU) in the Agile Mission Suite (AMS) Government Reference Architecture (GRA). The PCaB is a component of the Compute Cluster Farm (CCF) and is used to house other MPUs.

1.1 Purpose

The PCaB MPU Volume introduces the PCaB MPU and describes its key features, logical context, interfaces, and compliance requirements. The key features provide design guidance on what is typically expected for the PCaB MPU. The interface diagram contains a graphical representation of the types of interfaces that can exist (from the perspective of the PCaB MPU). The interfaces table contains a textual representation of the interfaces and their descriptions. Compliance requirements describe what is required for the PCaB MPU to achieve an AMS GRA compliance tier.

1.2 Scope

The PCaB volume supports compliance assessments for AMS GRA adopters.

1.3 Authoritative References

Authoritative references are key references that a procuring authority, Mission System Integrator (MSI), or MPU provider refers to when determining whether a given AMS GRA entity is compliant with AMS GRA business and architecture requirements. An AMS GRA entity is compliant when it provides the required prescriptive constructs (e.g. interfaces for an MPU). See **Table 1.3.1** for a list of authoritative references for the PCaB Volume.

Standard	Title	Version/Date
AMS GRA Volumes	Compliance Assessment Approach Volume	14.0
AMS GRA Volumes	Mission System Volume	14.0
AMS GRA Volumes	PCaB Compliance Documentation Template	14.0
AMS GRA Models	AMS GRA Architecture Model	14.0

Table 1.3.1: Authoritative References Identify the Key Documents Necessary for Performing a Compliance Assessment of the PCaB.

1.4 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or MPU provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.4.1** for a list of informative references for the PCaB Volume.

Standard	Title	Version/Date
AMS GRA Volumes	DPP MPU Volume	14.0
AMS GRA Volumes	Switch MPU Volume	14.0
AMS GRA Volumes	Data Storage MPU Volume	14.0
AMS GRA Volumes	CMM MPU Volume	14.0
AMS GRA Volumes	CDS MPU Volume	14.0
AMS GRA Volumes	Cryptographic Unit MPU Volume	14.0
AMS GRA Volumes	Time Distribution Volume	14.0

Table 1.4.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About the PCaB and Related Entities.

2.0 Processor Chassis and Backplane (PCaB)

The PCaB MPU installs into the platform and may host the following MPUs:

- Countermeasures Manager (CMM)
- Digital Payload Processor (DPP)
- Switch
- Cross Domain Solution (CDS)
- Cryptographic Unit
- Data Storage MPUs

The PCaB MPU interfaces with the platform (mechanical, cooling/thermal, power, and network) infrastructure. The PCaB MPU provides local network interconnections between hosted MPUs and the platform networks. PCaBs that host any of the aforementioned supported hardware are referred to as populated PCaBs. A populated PCaB does not have any additional compliance requirements and should not be equated with any layer of the AMS GRA architecture, (e.g. Multi-Function Processor (MFP)), because it can host MPUs which reside at any of the levels.

2.1 Key Features

Table 2.1.1 identifies the key features of PCaBs and the benefit of these features. This table is not prescriptive but identifies a number of attributes that AMS GRA definitions and interfaces are designed to support. This table is included in the PCaB documentation template so that PCaB providers can summarize the specific features supported by a PCaB instantiation.

Key Feature	Benefit
Backplane	The PCaB MPU contains a backplane. A backplane is a collection of static electrical and/or optical interconnections between the slots of the chassis. The backplane provides interconnections for the system management bus and special signals.
Chassis	The PCaB MPU contains a chassis. A chassis is a physical frame that contains the other hosted MPUs which may be installed into the platform. The chassis provides mechanical, electrical, optical, and thermal connections between the chassis and the platform. The chassis adheres to the standards specified by the platform for the chassis-to-platform interfaces and for the internal chassis-to-hosted MPU interfaces.

Key Feature	Benefit
Cooling	The PCaB MPU contains cooling/thermal infrastructure. The cooling/thermal infrastructure interfaces with the platform cooling/thermal infrastructure and provides the cooling/thermal management of the MPUs hosted within the chassis.
Mounting	The PCaB MPU contains mounting infrastructure. Externally, this mounting infrastructure interfaces with the mechanical mounting infrastructure of the platform to enable the PCaB MPU to mount within the platform. Internally, the PCaB MPU mounting infrastructure interfaces with the MPUs hosted within the PCaB MPU enabling secure mounting within the PCaB MPU.
PCaB System Management	The PCaB MPU optionally contains a system management function. The system management function oversees the power state, and health and status of the PCaB MPU and the MPUs hosted within the PCaB MPU. The system management function interconnects with the hosted MPUs via the system management bus interconnects resident in the backplane.
Power Supply	The PCaB MPU contains a power supply. The power supply interfaces with the power infrastructure of the platform. The power supply provides power to the slot where hosted MPUs reside within the chassis.

Table 2.1.1: The PCaB Key Features Table Describes the Specific Functionality Supported by a PCaB Instantiation.

2.2 Interface Diagram

Figure 2.2.1 depicts the PCaB MPU and its external interfaces. The interface diagram only contains logical entities.

The Multi-Level Security (MLS) architecture of AMS GRA drives the PCaB interface definitions when physical separation is required. The interface definitions of hosted Data Storage, Switch, CDS, Cryptographic Unit, DPP, and CMM MPUs drive the MLS architecture and influence the design of the PCaB.

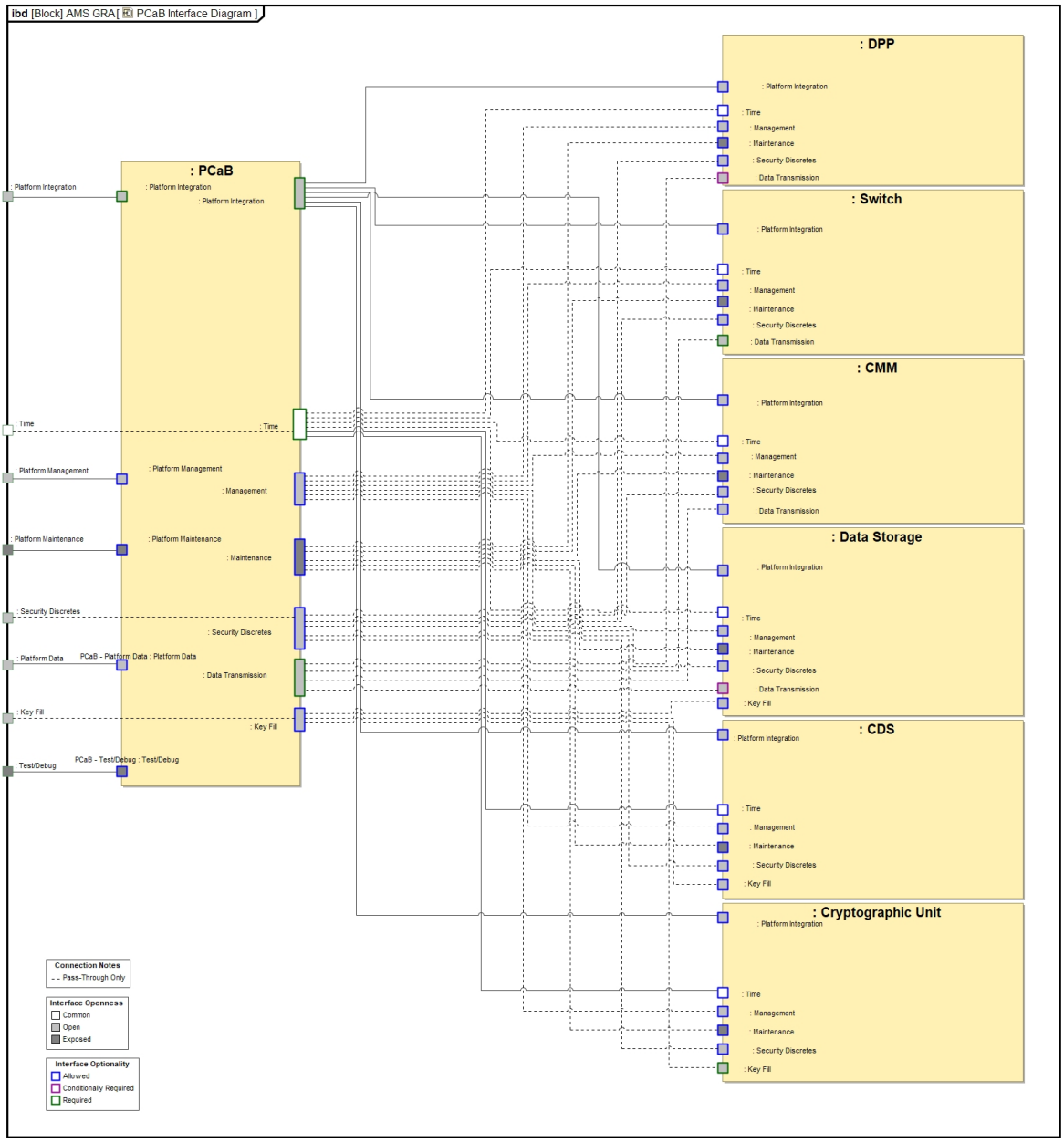


Figure 2.2.1: This PCaB Interface Diagram Shows Logical Interfaces of the PCaB and Their Connections with Other AMS GRA MPUs.

2.3 Interfaces Table

Table 2.3.1 identifies the PCaB Interfaces depicted in **Figure 2.2.1**. Note that there are two different Platform Integration interfaces on the PCaB representing two different relationships in **Figure 2.2.1**. The first Platform Integration relationship runs between the host platform and the PCaB, which represents integration with the host platform. The second Platform Integration relationship runs between the PCaB and other MPUs. This relationship enables integration with the MPUs.

When determining whether a given MPU or infrastructure is AMS GRA compliant, Interface Optionality is defined as follows:

- Required interfaces must be implemented
- Conditionally Required interfaces must be implemented when specific conditions are satisfied
- Allowed interfaces are optional to implement

Interface	Documentation	Interface Optionality
Data Transmission	The Data Transmission interface represents the interface support used to facilitate a data transmission connection. The interface is not concerned with what type of data is being transmitted, it only denotes what MPUs can be connected for transmitting data.	Required
Key Fill	The Key Fill interface is used to load keys into the system to support encryption across the architecture. This is a dedicated path to load a master key or keys that then support key derivations to be used by other MPUs to encrypt communication or for other cryptographic functions.	Allowed
Maintenance	The Maintenance interface provides the capability to perform preventative and corrective maintenance on installed PCaB modules.	Allowed
Management	The Management interface provides the capability to support management of the MPU in an AMS GRA compliant mission system. Example management for the PCaB includes centralized control and monitoring (e.g. Intelligent Platform Management Interface (IPMI)) of PCaB modules.	Allowed
Platform Data	The Data-to/from-platform interface specifies the data interfaces between the PCaB and the platform and other PCaBs. The platform includes the platform mission processing and vehicle management processing.	Allowed

Interface	Documentation	Interface Optionality
Platform Integration	The Platform Integration interface connecting PCaB to the internal component MPU represents the general interfaces that a PCaB will typically have with the MPUs it may host, including but not limited to mounting, electrical, hydraulic, and cooling connections.	Required
Platform Integration	The Platform Integration interface connecting the PCaB MPU to the host platform represents the general interfaces that a PCaB will typically have with the host platform, including but not limited to mounting, electrical, hydraulic, and cooling connections.	Required
Platform Maintenance	The Platform Maintenance interface provides the capability to perform preventative and corrective maintenance while installed on the platform.	Allowed
Platform Management	The Platform Management interface provides the capability to support management of the MPU in an AMS GRA compliant mission system. Example management for the PCaB includes centralized control and monitoring (e.g. IPMI).	Allowed
Security Discretes	The Security Discrete specifies the security interfaces between the platform and the PCaB as well as between the PCaB modules and the platform. Security discretes may include a zeroize discrete and a write protect/non-volatile read-only discrete.	Allowed
Test/Debug	The Test/Debug interface specifies an interface to use for test equipment purposes.	Allowed
Time	The Time interface specifies the time sync interface between the PCaB and the PCaB modules.	Required

Table 2.3.1: The Procuring Authority, MPU Provider, and MSI Use this Table to Identify the Interfaces to Implement for a Compliant MPU or Infrastructure.

3.0 Compliance Requirements

Table 3.0.1 states the applicable MPU compliance requirements for the PCaB MPU.

See the Compliance Assessment Approach Volume for more information on the tiered requirement strategy as well as the definitions of the interface characteristics.

MPU developers need to coordinate with the procuring authority and applicable MSI to receive additional guidance beyond basic compliance. This coordination includes architectural guidance to ensure the MPU integrates within the overall Mission System architecture as well as guidance to align with Commercial-Off-The-Shelf (COTS) standards such as physical form factor, cooling strategy, and environmental operating conditions.

See the Mission System Volume for additional details about MPU Fault Reporting.

Note: The verification methods listed in the requirements table are AMS GRA recommendations. An adopting program may alter the recommended verification methods In Accordance With (IAW) the needs of the program. Refer to the Compliance Assessment Approach Volume for more information.

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-001	An MPU shall be documented IAW the corresponding AMS GRA compliance documentation template or contract template, as applicable.	Tier 1	Inspection	
GRA-MPU-025	An MPU shall report faults with the associated fault status.	Tier 1	Inspection	Off-nominal behavior detailed in compliance documentation.
GRA-MPU-026	An MPU shall implement Required interfaces specified IAW the AMS GRA Model.	Tier 1	Inspection	
GRA-MPU-027	An MPU shall implement all interfaces with the level of Openness specified IAW the AMS GRA Model.	Tier 1	Analysis	
GRA-MPU-029	An MPU shall implement only the interfaces in the corresponding MPU interface table.	Tier 2	Inspection	
GRA-MPU-030	An MPU shall document interface openness IAW the Compliance Assessment Approach.	Tier 1	Inspection	
GRA-MPU-031	An MPU shall include a model compatible with MBSE content contained in the AMS GRA SK release with which the MPU is compliant.	Tier 2	Inspection	
GRA-MPU-032	An MPU shall include a model that identifies all external MPU interfaces.	Tier 2	Inspection	
GRA-MPU-038	An MPU shall use open, commercial standards for the Management interface.	Tier 1	Inspection	
GRA-MPU-041	An MPU shall document SAST results for all microprocessor software source code in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	
GRA-MPU-042	An MPU shall document DAST results for all microprocessor software in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	
GRA-MPU-046	An MPU shall provide automated tests of non-COTS microprocessor software with at least 50% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 1	Demonstration	Only applicable to MPUs that include non-COTS microprocessor code.
GRA-MPU-059	An MPU that decrypts and/or encrypts classified data and/or CPI shall only store classified plain text keys in volatile memory.	Tier 1	Inspection	
GRA-MPU-061	An MPU shall ensure the authenticity and integrity of MPU software and firmware during MPU initialization.	Tier 1	Inspection	

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-064	An MPU shall use NSA-approved algorithms to authenticate and verify the integrity of a software load when received over a non-secure communication channel.	Tier 1	Demonstration	
GRA-MPU-065	An MPU shall use NSA-approved algorithms to authenticate and verify the integrity of software locally stored at rest, not including bootloaders; prior to updating, loading, and executing.	Tier 1	Inspection	
GRA-MPU-066	An MPU shall support the use of a Mission System-provided RoT.	Tier 1	Demonstration	
GRA-MPU-077	An MPU shall provide automated tests of non-COTS microprocessor software with at least 80% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 3	Demonstration	
GRA-MPU-078	An MPU model shall be capable of automated verification with AMS GRA models.	Tier 3	Inspection	
GRA-MPU-079	An MPU shall include a model that provides definition of all external MPU interfaces.	Tier 3	Inspection	
GRA-MPU-084	An MPU that implements the Time interface shall use one or more of PTP, and OMS legacy timing.	Tier 1	Inspection	

GRA-MPU-085	An MPU shall report faults with the associated fault status using the applicable interface. (See notes).	Tier 2	Inspection	Cryptographic Unit MPUs and CDS MPUs report faults as defined in the CDS and CU IDD. Local Function MPUs report faults as defined in Local Function DK. MFA MPUs report faults as defined in the applicable MEL IDD. AMS GRA Service, CMM, DPP, Data Storage, Switch, OMS Service, and PCaB MPUs report faults using a software interface in UCI format.
GRA-MPU-091	An MPU shall report BIT information on its Management Interface.	Tier 1	Inspection	Note: MFA reports BIT information via the MEL interface

Table 3.0.1: A PCaB Volume's Adherence to the Compliance Requirements Determines Its Compliance Score. See the Compliance Assessment Approach Volume for More Details.

A.0 Acronyms

AMS	Agile Mission Suite
BIT	Built-In Test
CCF	Compute Cluster Farm
CDS	Cross Domain Solution
CMM	Countermeasures Manager
COTS	Commercial-Off-The-Shelf
CPI	Critical Program Information
CU	Cryptographic Unit
DAST	Dynamic Application Security Testing
DK	Development Kit
DPP	Digital Payload Processor
GRA	Government Reference Architecture
IAW	In Accordance With
IDD	Interface Description Document
IPMI	Intelligent Platform Management Interface
MBSE	Model-Based Systems Engineering
MEL	Multi-Function Aperture (MFA) Encapsulation Layer
MFA	Multi-Function Aperture
MFP	Multi-Function Processor
MLS	Multi-Level Security
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
NSA	National Security Agency
OMS	Open Mission Systems
PCaB	Processor Chassis and Backplane
PTP	Precision Time Protocol
RoT	Root of Trust

SAST	Static Application Security Testing
SK	Starter Kit
UCI	Universal Command and Control Interface

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revision did not introduce any significant changes.	July 24,2025	SK13.0
B	This revision did not introduce any significant prose changes. Adds new requirement GRA-MPU-091	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the PCaB Volume Between Documentation Releases.